

- 5 (a) (i) region of space which a force is felt/experienced B1
- (ii) explanation that relates field strength to the closeness of lines B1
- (b) force on the magnet is downward B1
- force on current upwards by Newton's third law M1
- magnetic field directed towards P, P is South A1
- (c) (i) arrow from below foil to above the foil B1
- (ii) centripetal force of charged particle in circular motion provided by magnetic force
- $$\frac{mv^2}{r} = Bqv \Rightarrow mv = Bqr \Rightarrow \text{momentum} \propto \text{radius} \quad \text{C1}$$
- ratio = $5.7 / 7.4 = 0.77$ A1

